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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2016

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

(2013–2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. What is the purpose of Accumulator?
2. What is the purpose of Memory Buffer Register (MBR)?
3. What is an instruction?
4. State the need for subroutines.
5. Define : interface.
6. What is DMA?

7. State the basic operation of cache memory.
8. What is RAM?
9. What is meant by hardwired control?
10. Why the data bus is bidirectional?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. What is meant by addressing mode? Explain the different types addressing modes in detail.

Or

12. Explain about shift and rotate instructions in detail.

UNIT II

13. Explain about IA-32 instructions in detail.

Or

14. Discuss about IA-32 assembly language with examples.

UNIT III

15. Discuss about Pentium interrupt structure in detail.

Or

16. Explain about handling multiple devices in detail.

UNIT IV

17. What is virtual memory? Explain in detail.

Or

18. Explain about semiconductor RAM memories in detail.

UNIT V

19. Explain how the instructions are executed in a computer detail.

Or

20. What is meant by pipelining? Discuss in detail.
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B.Tech. DEGREE EXAMINATION, NOVEMBER 2013.

Fourth Semester

Computer Science and Engineering

COMPUTER ORGANISATIONS AND
ARCHITECTURE

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. List the different types of computer registers.
2. What is accumulator logic?
3. Define program counter.
4. Define routine.
5. Define hit ratio.
6. Define virtual memory.
7. Define baud rate.

8. What is the difference between synchronized and asynchronous serial transfer.

9. What do you mean by symmetric multi processor?

10. Define vector computation.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions. ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the different phases of instruction cycle.

Or

12. Explain timing and control concept.

UNIT II

13. Give a detailed account on instruction formats.

Or

14. Explain division algorithm.

UNIT III

15. Explain auxiliary and cache memory.

Or

16. Explain the virtual memory concept.

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UNIT IV

17. Explain the various categories of message framing techniques.

Or

18. Explain asynchronous data transfer.

UNIT V

19. Explain about the multiple processor organizations and its types.

Or

20. Explain cache coherence.

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B.Tech. DEGREE EXAMINATION,
MAY 2018.

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE
(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is an instruction cycle? Give examples of instructions with different number of instruction cycles.
2. List the different types of computer registers.
3. What is program control?
4. Differentiate between floating point and decimal arithmetic operations.
5. What is the use of Cache memory?
6. Why Virtual memory is required?
7. What are the functions of input output interface?

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8. Distinguish between the different modes of data transfer.
9. What are the different types of multiple processor organizations?
10. What are the advantages of non-uniform memory access?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.
All questions carry equal marks.

UNIT I

11. Describe different types of computer instructions with examples for each.

Or

12. Explain the design of basic computer and accumulator logic.

UNIT II

13. What are the different addressing modes? Explain with examples.

Or

14. With suitable diagrams explain register organization and stack organization.

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UNIT III

15. Explain any three types of memory.

Or

16. Compare and contrast different types of memory.

UNIT IV

17. What is cycle stealing? Explain how they are used in any two instances in the working of a computer.

Or

18. What are the different types of interrupt? With suitable examples explain how they are used in both system and application programs.

UNIT V

19. Explain how Cache coherence could be achieved.

Or

20. Describe Vector Computation.

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UNIT V

19. Consider un-pipelined machine with 10 ns clock cycles. It uses 4 cycles for ALU operations and branches whereas five cycles for memory operations. Assume that the relative frequencies of these operations are 40%, 20% and 40% respectively. Suppose that due to clock skew and setup, pipelining the machine adds 1 ns overhead to the clock. Calculate gain in the instruction execution rate from a pipeline.

Or

20. Discuss the following terms :
- (a) Cache structure in Pentium
 - (b) Brief Super scalar architecture
 - (c) What are the advantages of hardwired control unit? What are the disadvantages?

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Third Semester

Computer Science and Engineering
COMPUTER ORGANIZATION AND
ARCHITECTURE

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Brief zero address instructions format.
2. Define the fundamental conceptual unit in a computer.
3. Compare the features of P-V and P-VI in brief.
4. Distinguish I/O instructions and I/O commands.
5. The interrupt vector table is always created in the first 1 K area of the memory. Justify the statement.
6. Define interface circuit. List any two I/O interfaces and its purpose.
7. Briefly explain the significance of DDRAM.

8. Explain the need of overlaying concept in memory management.
9. Distinguish RISC and CISC architecture.
10. Brief data hazards and instruction hazards in pipelining techniques.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Define subroutine. Explain subroutine linkage using a link register. (6)
- (b) Brief (i) byte addressability (ii) big-endian assignment (iii) little-endian assignment. (5)

Or

12. (a) With a neat sketch, explain single bus and multiple bus structure used to inter-connect functional units in computer system. (6)
- (b) Explain how the performance of a computer can be measured. List the measures that are used to improve the performance of computers. (5)

UNIT II

13. (a) What is memory paging? Explain how it is used for memory addressing in Pentium processor. (6)
- (b) Write an assembly language program to find one's complement and two's complement of an 8-bit number. (5)

Or

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14. (a) What features have been added to P-IV to improve Paging? (5)
- (b) Find greatest of five numbers and the GCD for the same five numbers using assembly language program. (6)

UNIT III

15. Explain about the following Input-Output interface with an example : (6)
- (a) I/O bus and interface modules
- (b) Isolated versus memory mapped I/O
- (c) I/O versus memory bus.

Or

16. With a neat sketch, explain the Pentium interrupt structures.

UNIT IV

17. Explain the following memory organization methods in virtual memory with diagrams : (6)
- (a) Address space and memory space
- (b) Address mapping using pages
- (c) Associative memory page table
- (d) Three benefits of virtual memory

Or

18. (a) Explain in detail the various memory management techniques. (6)
- (b) Explain the various modes in which DMAC works. (5)

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2016.

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Bring the difference between computer architecture and computer organization.
2. Define multiprocessor.
3. Write notes on IA-32 Pentium.
4. Give the instruction format of IA-32.
5. Draw the I/O interface for an input device.
6. Define distributed arbitration.
7. Define asynchronous DRAMs.
8. What are the types of flash memory?
9. Define multiphase clocking.
10. What are the steps involved in processing the instruction in pipelined processor?

PART B — (5 × 11 = 55 marks)

Answer ALL the questions.

UNIT I

11. Explain the basic instruction types.

Or

12. With an example, explain indirection and pointers.

UNIT II

13. Discuss various addressing modes of IA-32.

Or

14. Explain subroutines concept in IA-32 architecture.

UNIT III

15. Write notes on :

(a) Enabling and disabling interrupts (5)

(b) Handling multiple devices. (6)

Or

16. Discuss about :

(a) Pentium interrupt structure (6)

(b) Direct memory access. (5)

UNIT IV

17. Explain about mapping functions in cache memory.

Or

18. Explain address translation in virtual memory.

UNIT V

19. Explain hardwired control in detail.

Or

20. Explain conditional branches and branch prediction in detail.

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2014.

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. Differentiate computer organization and computer architecture.
2. List the four phases of instruction cycle.
3. Mention the purpose of program counter and accumulator.
4. What are the main features of Booth's algorithm?
5. What are the advantages of using virtual memory?

6. Define latency time.
7. What is polling?
8. What do you mean by memory mapped I/O?
9. What are the advantages of symmetric multi-processors?
10. What is meant by uniform memory access?

PART B — (5 × 11 = 55 marks)

Answer ALL the questions. ONE from each Unit.

All questions carry equal marks.

UNIT - I

11. Explain the various instruction types with example.

Or

12. Discuss the processor level design of basic computer with neat diagram.

UNIT - II

13. Describe the non-restoring division algorithm with example.

Or

14. Define addressing mode. Classify addressing modes and explain each type with example.

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UNIT - III

15. Explain the various mapping techniques associated with cache memories.

Or

16. (a) Explain the address translation method in virtual memory.
(b) Write a short note on magnetic hard disks.

UNIT - IV

17. What is DMA? Draw the block diagram and explain it in detail.

Or

18. Describe the vector interrupt scheme with a neat block diagram.

UNIT - V

19. Illustrate the taxonomy of parallel processor architectures.

Or

20. Define cache coherence. Describe the MESI state transition diagram in detail.

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B.Tech. DEGREE EXAMINATION, APRIL 2013.

Fourth Semester

Computer Science and Engineering

COMPUTER ORGANISATIONS AND
ARCHITECTURE

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is program counter?
2. Comment on instruction format types.
3. What is micro program?
4. Write the methods used for dividing numbers.
5. What is boots trap loader?
6. Write the difference in between pages and blocks.
7. List out modes of data transfer to and from peripherals.

8. Comment on IOP.

9. What is NUMA?

10. Comment on RWITM.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, ONE from each unit.

All questions carry equal marks.

UNIT I

11. Discuss about input-output and interrupt.

Or

12. Explain the memory reference instructions.

UNIT II

13. Discuss about instruction formats and addressing modes.

Or

14. Explain the algorithm for multiplication.

UNIT III

15. Discuss about associate memory organization in detail.

Or

16. Explain the virtual memory organization in detail.

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UNIT IV

17. Describe the interrupt priority in detail.

Or

18. Discuss about serial communication in I/O organization.

UNIT V

19. Explain about cache coherence and MESI protocol.

Or

20. Discuss about vector computation.

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2015.

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What are Multiprocessors?
2. Define Addressing mode? What are subroutines?
3. What are the Registers available with IA-32?
4. Give example for Shift/Rotate instructions?
5. What do you mean by an interrupter?
6. What are the advantages of Direct Memory Access?

7. What are the various types of ROM?

8. Define virtual memory.

9. Write differences between Hardwired and microprogrammed controls.

10. How can we solve data Hazards in pipelining?

PART B — (5 × 11 = 55 marks)

Answer ALL the questions, ONE from each unit.

All questions carry equal marks.

UNIT I

11. Explain the various types of addressing mode with example.

Or

12. Explain the shift and rotate operations with suitable example.

UNIT II

3. Explain the architecture of IA-32 with suitable Block diagram.

Or

Explain the I/O operations IA-32 with example.

UNIT III

15. Explain the interrupt enabling and disabling mechanism.

Or

16. Explain the process of DMA with suitable diagram.

UNIT IV

17. Explain the memory hierarchy with respect to speed, size and cost.

Or

18. Explain the concept of virtual memories.

UNIT V

19. Give note on

(a) Complete Instruction Execution (6)

(b) Microprogrammed control. (5)

Or

20. What are data and Instruction hazards and how this can be avoided?